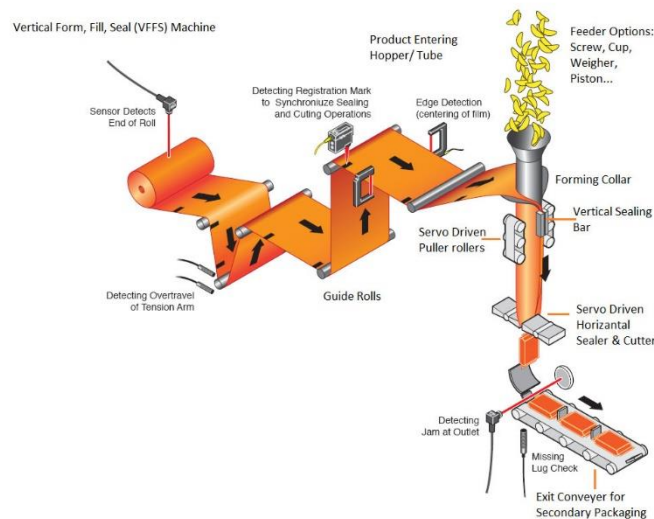


1. Introduction

1.1 Overview of Vertical Form-Fill-Seal (VFFS) Machinery

The Vertical Form-Fill-Seal (VFFS) packaging system represents one of the most ubiquitously deployed automated technologies within contemporary packaging engineering. This integrated machinery operates on a consolidated principle that synergizes three fundamental processes—forming, filling, and sealing—into a single, continuous production line.

The operational sequence commences with the extraction of thermoplastic film from a supply reel, which subsequently traverses a forming collar (forming tube) that metamorphoses the planar film into a cylindrical configuration. Longitudinal sealing bars execute a continuous thermal welding of the film's lateral edges, thereby generating a seamless tubular structure. Concurrently, transverse sealing jaws perform intermittent sealing operations to create the pouch's bottom seal and execute its separation post-filling. This system exhibits remarkable versatility, accommodating diverse pouch dimensions and a broad spectrum of packaging materials, rendering it an optimal solution for high- to medium-volume production environments.



[Fig. 1.1: Schematic representation of VFFS operational sequence]

1.2 Project Objectives

The principal aim of this undertaking is the conception and analysis of an advanced packaging system for VFFS machinery that satisfies the following operational specifications:

Primary Objective: To develop a packaging system capable of producing 12–30 pouches per minute, with a constant unit weight of 500 grams per package, whilst maintaining weight accuracy within a tolerance of $\pm 2\%$.

Secondary Objectives:

- Attainment of elevated operational efficacy through the minimization of packaging material wastage to below 1%
- Assurance of thermal sealing integrity with a minimum weld strength of 20 Newtons per 15 millimeters
- Implementation of an adjustable architecture compatible with variable pouch widths ranging from 80 to 150 millimeters
- Achievement of mechanical stability at elevated operational velocities with a safety factor not less than 2.5

Fig. 1.2: Target Performance Parameters Matrix

Category	Parameter	Target Specification	Required Standard
Productivity	Output Rate (\$N_b\$ bags)	12 – 30 pouches/min	Based on v_{max}
Weight Control	Unit Weight	500 grams (constant)	
			Gravimetric Precision
Accuracy	Weight Tolerance	$\pm 2\%$	Quality Control
Material Usage	Wastage Rate	Material Efficiency	
Seal Integrity	Minimum Weld Strength	$\Delta \in 20N/15mm$	Thermal Integrity
Flexibility	Pouch Width (w)	80 – 150 mm	Mechanical Stability
Safety	Pouch Width (w)	Structural Stability	
$\Delta \in 2.5$	Mechanical Safety Factor (SF)		

[Fig. 1.2: Target performance parameters matrix]

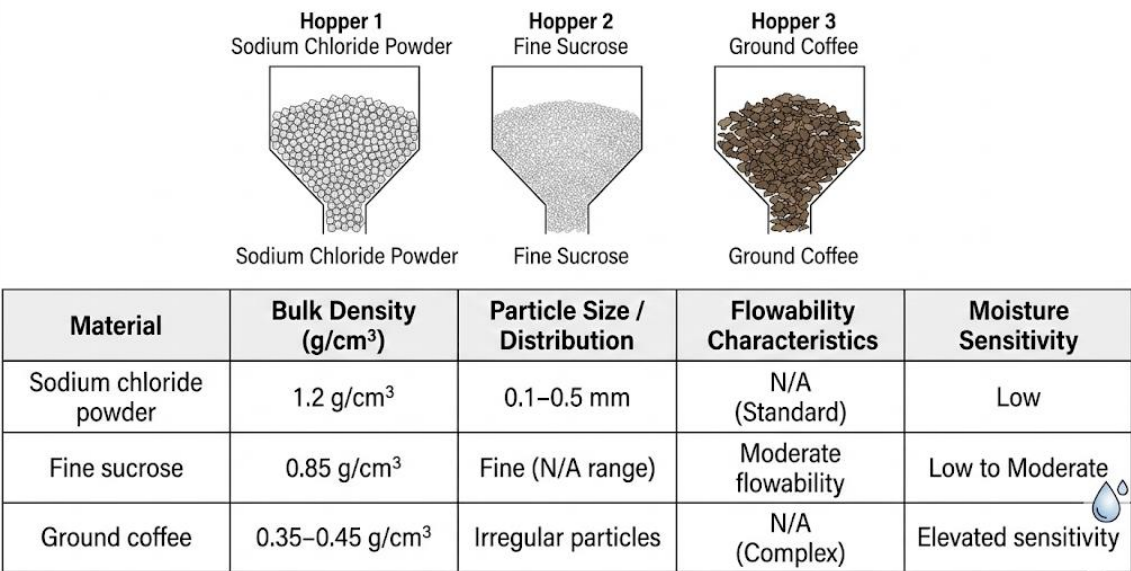
1.3 Scope of Work

This project concentrates on the packaging of dry granular and pulverulent materials, specifically:

Target Materials:

- **Sodium chloride powder:** Bulk density 1.2 g/cm³, particle size distribution 0.1–0.5 mm
- **Fine sucrose:** Bulk density 0.85 g/cm³, moderate flowability characteristics
- **Ground coffee:** Bulk density 0.35–0.45 g/cm³, elevated moisture sensitivity

The scope encompasses the design of all constituent components within the packaging system, commencing from the film unwinding mechanism through to the final pouch cutting apparatus, with particular emphasis on adherence to food packaging regulatory standards. Explicitly excluded from this scope are electronic control systems and external material handling conveyors; the investigation is strictly confined to the mechanical and structural facets of the packaging apparatus.



⚠️ *Specific range data not available for all parameters; values are typical or qualitative based on specified characteristics.*

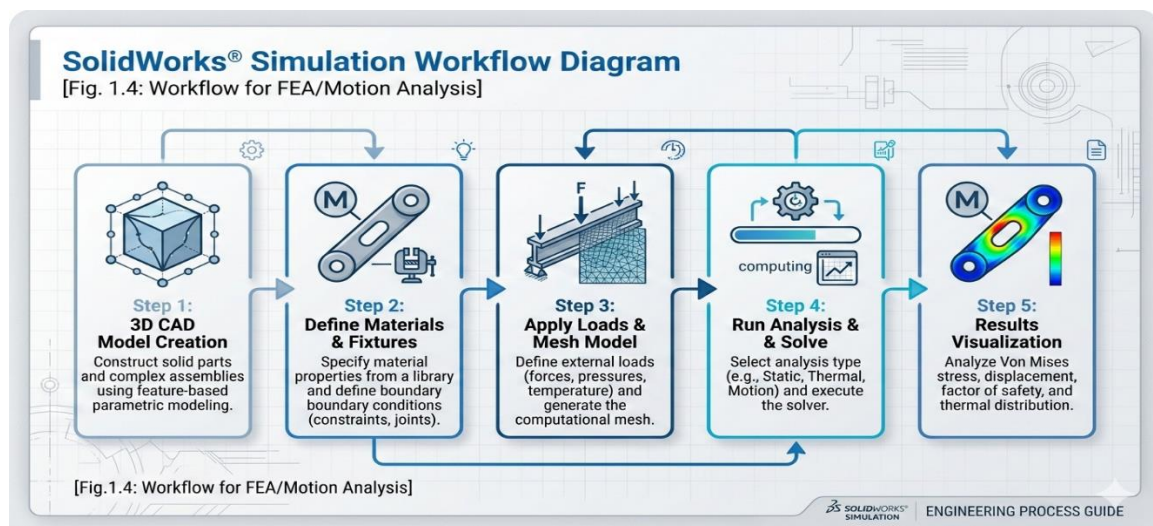
[Fig. 1.3: Material properties comparison chart]

1.4 Utilized Software and Computational Tools

SolidWorks® – Primary Engineering Platform

SolidWorks® has been selected as the principal software environment for executing all phases of engineering design and analysis within this project, predicated upon the following capabilities:

- **Advanced Three-Dimensional Modeling:** The platform provides a comprehensive ecosystem for parametric solid modeling and complex assembly management, facilitating the precise design of all packaging system components with sophisticated interference detection and kinematic simulation capabilities.
- **Integrated Finite Element Analysis:** The embedded SolidWorks Simulation® suite enables comprehensive static stress analysis, thermal analysis, and motion analysis to be conducted on the unified model architecture, thereby eliminating data translation inefficiencies between disparate software packages.
- **Manufacturability Integration:** The software supports the direct generation of manufacturing documentation and computer-aided manufacturing (CAM) data, thus streamlining the transition from conceptual design to physical prototyping and production.



[Fig. 1.4: SolidWorks Simulation workflow diagram]

2. Theoretical Design and Preliminary Analysis

2.1.1 Film Motion Equations

The film movement in the VFFS system is governed by a comprehensive set of fundamental forces and equations:

Film Linear Velocity Equation:

$$v_f = \frac{L_{bag} \times N_{bags}}{60}$$

Where:

- v_f = Film velocity $\left(\frac{m}{s}\right)$
- L_{bag} = Bag length = 0.25 m (for 500g package)
- N_{bags} = Number of bags/minute = 12 – 30 bags/minute

For target productivity:

- Minimum velocity: $v_{min} = \frac{12 \times 0.25}{60} = 0.05 \text{ m/s}$
- Maximum velocity: $v_{max} = \frac{12 \times 0.25}{60} = 0.125 \text{ m/s}$

Acceleration and Deceleration Equation:

$$a = \frac{v^2 - v_0^2}{2s}$$

Where:

- a = Acceleration $\left(\frac{m}{s^2}\right)$
- v_0 = Initial velocity = 0
- s = Acceleration distance = 0.05 m

Required Film Pulling Force:

$$F_{pull} = m \cdot a + F_{friction} + F_{tension}$$

Where:

- $F_{friction} = \mu \times N$ (Friction force)
- μ = Coefficient of friction = 0.3 for plastic film
- $F_{tension}$ = Required tension force = 50 – 100 N

[Fig. 2.1: Free body diagram illustrating forces acting on film]

2.1.2 Tension and Sealing Force Calculations

Film Tensile Stress:

$$\sigma = \frac{F}{A} = \frac{F}{t \times w}$$

Where:

- $\sigma = \text{Stress} \left(\frac{N}{mm^2} \right)$
- $F = \text{Tension force (N)}$
- $t = \text{Film thickness} = 0.06 \text{ mm}$
- $w = \text{Film width} = 320 \text{ mm}$

Maximum Allowable Stress:

$$Q_{max} = \frac{\sigma_{yield}}{SF}$$

Where:

- $\sigma_{yield} = \text{Polyethylene yield stress} = 20 - 30 \text{ MPa}$
- $SF = \text{Safety factor} = 2.5$

Required Sealing Force:

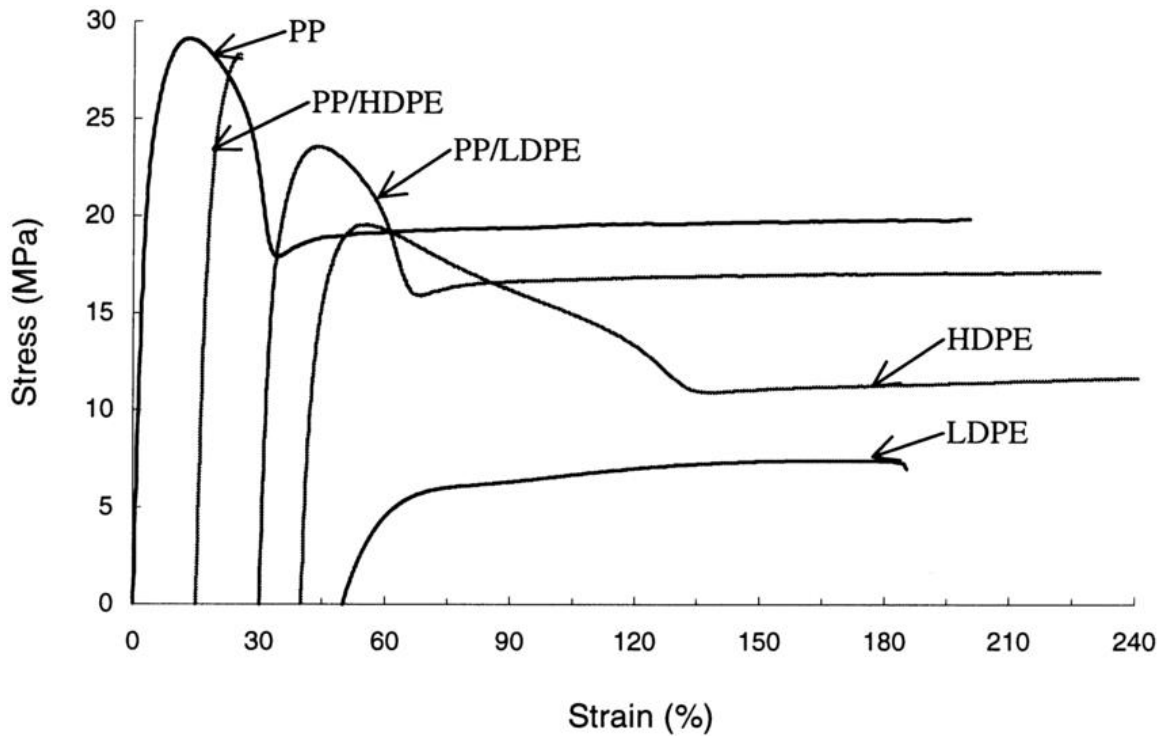
$$F_{seal} = P_{seal} \times A_{seal}$$

Where:

- $P_{seal} = \text{Sealing pressure} = 2 - 4 \text{ bar (200 - 400 kPa)}$
- $A_{seal} = \text{Sealing area} = \text{Jaw length} \times \text{Seal width} = 150 \times 5 = 750 \text{ mm}^2$

Actual force calculation:

$$F_{seal} = 400 \times 103 \times 750 \times 10^{-6} = 300 \text{ N}$$



[Fig. 2.2: Stress-strain curve for polyethylene material]

2.1.3 Heat Transfer Equations for Sealing

Conductive Heat Transfer Equation:

$$Q = \frac{k \times A \times \Delta T}{d} \times t$$

Where:

- Q = Heat transferred (J)
- k = Film thermal conductivity = $0.33 \text{ W}/(\text{m} \cdot \text{K})$
- A = Contact area = 750 mm^2
- ΔT = Temperature differential = $140^\circ\text{C} - 25^\circ\text{C} = 115^\circ\text{C}$
- d = Film thickness = 0.06 mm
- t = Sealing time

Required Sealing Time:

$$t_{seal} = \frac{Q_{required} \times d}{k \times A \times \Delta T}$$

Energy Required for Sealing:

$$E_{SEAL} = m \times C_p \times \Delta T$$

Where:

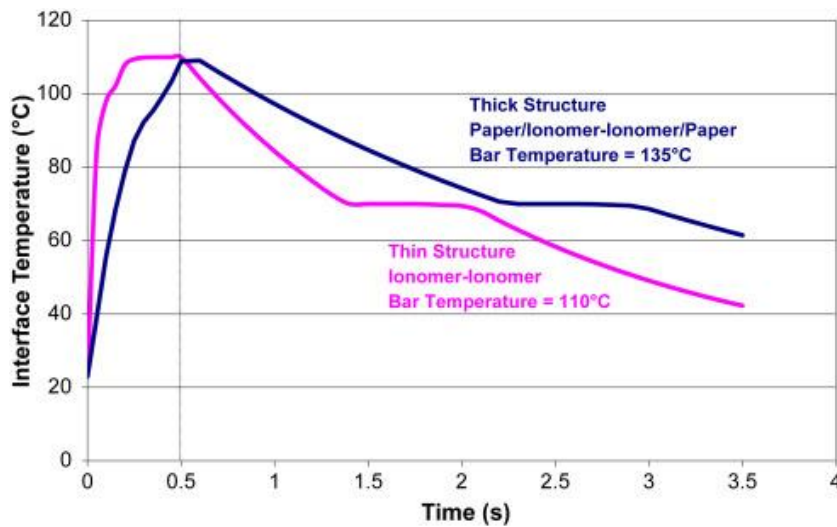
- $m = \text{Mass of sealing zone} = \rho \times V = 920 \times (750 \times 0.06 \times 10^{-9}) = 0.04 \text{ g}$
- $C_p = \text{Specific heat capacity of polyethylene} = 2.3 \text{ kJ}/(\text{kg} \cdot \text{K})$

Optimal Sealing Temperature:

$$T_{seal} = T_{melt} + 20^\circ\text{C}$$

Where:

- $T_{melt} = \text{Polyethylene melting point} = 120 - 130^\circ\text{C}$
- $T_{seal} = 140 - 150^\circ\text{C}$



[Fig. 2.3: Thermal sealing cycle diagram (Temperature vs Time)]

2.1.4 Speed and Productivity Calculations

Productivity Equation:

$$P = \frac{60}{t_{cycle}} \times \eta$$

Where:

- $P = \text{Productivity (bags/minute)}$
- $t_{cycle} = \text{Cycle time (seconds)}$
- $\eta = \text{System efficiency} = 0.85 - 0.95$

Cycle Time:

$$t_{cycle} = t_{pull} + t_{seal} + t_{cut} + t_{dwell}$$

Where:

- $t_{pull} = \text{Film pulling time} = L_{bag}/vf = 2 - 5 \text{ seconds}$
- $t_{seal} = \text{Sealing time} = 0.3 - 0.5 \text{ seconds}$
- $t_{cut} = \text{Cutting time} = 0.1 \text{ seconds}$
- $t_{cut} = \text{Dwell time} = 0.1 \text{ seconds}$

For target productivity of 30 *bags/minute*:

$$t_{cyclemax} = \frac{60}{30} = 2 \text{ seconds/bag}$$

Film Consumption Rate:

$$C_{flim} = N_{bags} \times (L_{bags} + L_{waste}) \times w_{film}$$

Where:

- $C_{flim} = \text{Film consumption (m}^2\text{/minute)}$
- $L_{waste} = \text{Waste length} = 0.01 \text{ m/bag}$
- $w_{film} = \text{Film width} = 0.32 \text{ m}$

Weight-based Productivity:

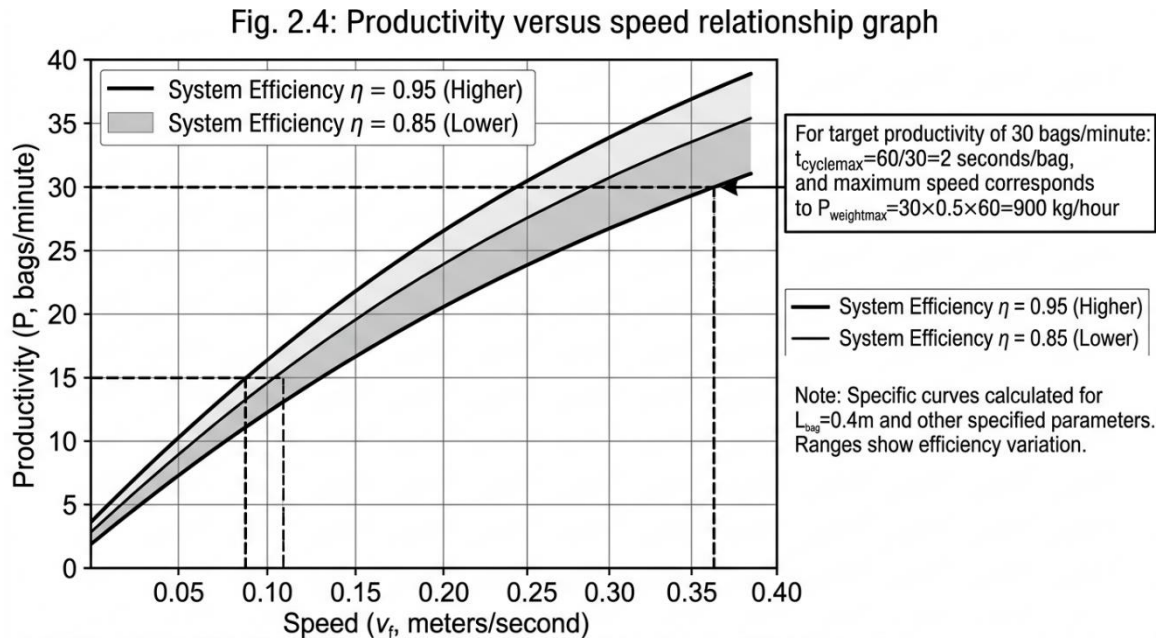
$$P_{weight} = N_{bags} \times m_{product} \times 60$$

Where:

- $P_{weight} = \text{Productivity (kg/hour)}$
- $m_{product} = \text{Product weight} = 0.5 \text{ kg}$

For maximum speed:

$$P_{weightmax} = 30 \times 0.5 \times 60 = 900 \text{ kg/hour}$$



[Fig. 2.4: Productivity versus speed relationship graph]

External Requirements List:

1. Images and Diagrams:

- Film force diagram: Technical illustration depicting forces acting on film
- Polyethylene stress-strain curve image
- Thermal sealing cycle chart (Temperature vs Time)
- Productivity versus speed graph

2. Reference Tables:

Material Thermal Conductivity (W/m·K) Melting Point (°C) Yield Stress (MPa)

<i>PE</i>	0.33	120 – 130	20 – 30
<i>PP</i>	0.22	160 – 170	30 – 40
<i>PET</i>	0.24	250 – 260	50 – 60

3. References and Resources:

- Book: "Flexible Packaging: Materials, Machinery, and Techniques" by F. Peter Boyle
- Standards: ISO 11607 for medical packaging systems
- Website: Packaging Digest for specifications
- Video: VFFS Machine Working Principle

4. Required Technical Data:

- Film specifications from supplier: thickness, tensile strength, elongation
- Product specifications: bulk density, angle of repose, particle size
- Sealing jaw catalog: dimensions, materials, temperature range

5. Software and Tools:

- MATLAB for differential equation solutions
- Excel for tabulation and graphical representations
- Engineering Equation Solver (EES) for thermal calculations

6. SolidWorks Images:

- Screenshot of film force analysis
- Thermal simulation image of sealing jaws
- Stress distribution graph

7. Standards and Regulations:

- ASTM D882: Tensile properties testing for plastic films
- ASTM F88: Seal strength testing
- FDA 21 CFR 177: Food contact material standard

2.2 Detailed Design

2.2.1 Forming Tube

Functional Description:

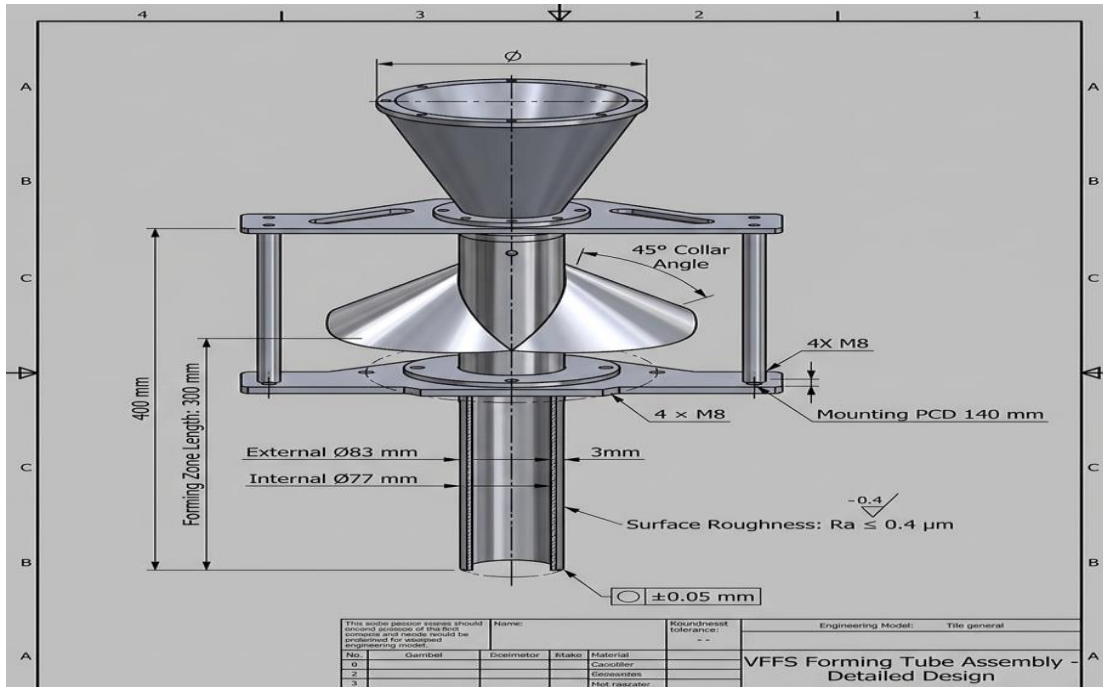
The forming tube constitutes the quintessential core component of the VFFS system, effectuating the transformation of planar film into a tubular configuration in preparation for product filling. The film traverses the external surface of the tube whilst the product flows through its internal cavity.

Design Specifications:

- External diameter: 83mm (for producing 120 mm width bags +10mm seal)
- Internal diameter: 77 mm (wall thickness 3 mm)
- Overall length: 400 mm
- Forming zone length: 300 mm
- Collar angle: 25° for optimal film folding fluidity

Engineering Specifications:

- Surface roughness: $R_a \leq 0.4 \mu\text{m}$ (mirror finish)
- Roundness tolerance: $\pm 0.05 \text{ mm}$
- Upper filling aperture: 75 – 80 mm diameter with mounting flange
- Mounting system: 4 × M8 threaded holes on 120 mm pitch circle diameter



[Fig. 2.5: Forming tube technical drawing with comprehensive dimensions]

2.2.2 Horizontal Sealing Jaws

Functional Description:

These components execute the creation of transverse upper and lower seals of the pouch whilst simultaneously severing the completed bag from the continuous film web.

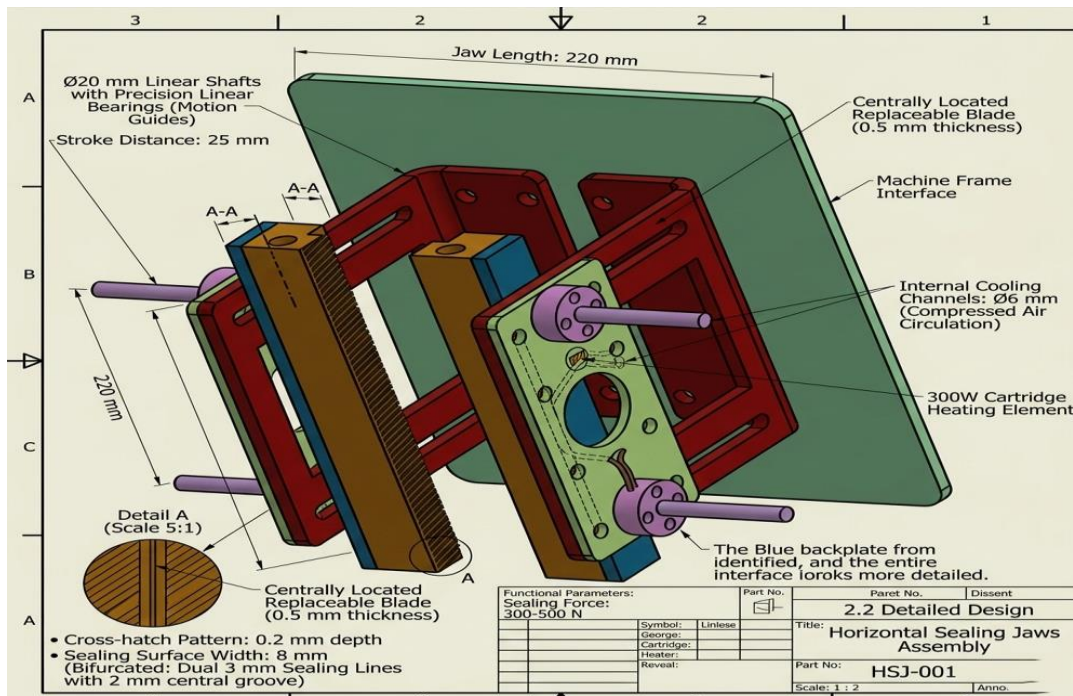
Design Specifications:

- Jaw length: 220 mm
- Sealing surface width: 8 mm (bifurcated into dual 3 mm sealing lines)
- Stroke distance: 25 mm
- Sealing force: 300-500 N
- Heating system: 300W cartridge heating element

Engineering Specifications:

- Sealing surface: Cross-hatch pattern with 0.2 mm depth
- Cutting mechanism: Centrally located replaceable blade (0.5 mm thickness)

- Cooling system: Internal channels Ø6 mm for compressed air circulation
- Motion guides: Ø20 mm linear shafts with precision linear bearings



[Fig. 2.6: Horizontal sealing jaw assembly cross-sectional view]

2.2.3 Vertical Sealing Bars

Functional Description:

These components effectuate the welding of overlapping film edges to form the continuous tubular structure (back seal).

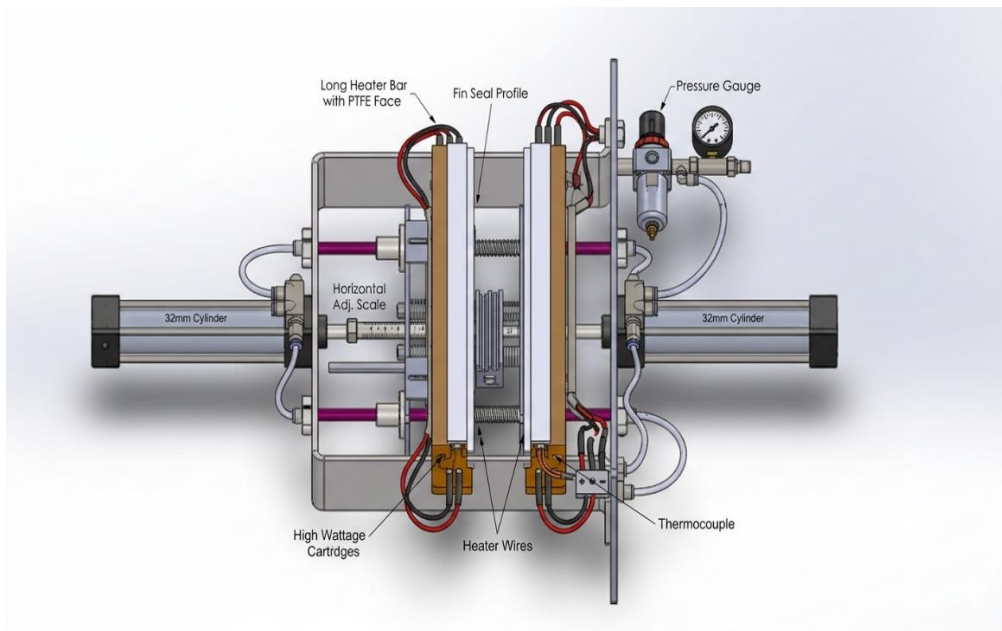
Design Specifications:

- Bar length: 400 mm
- Sealing surface width: 10 mm
- Pressure range: 3Bar
- Operating temperature: 140 – 160°C

Engineering Specifications:

- Seal configuration: Lap seal or fin seal arrangement
- Pressure mechanism: Ø32 mm pneumatic cylinder
- Position control: Adjustable ± 15 mm horizontally

- Thermal insulation: 3 mm PTFE layer



[Fig. 2.7: Vertical sealing mechanism schematic representation]

2.2.4 Film Pull Mechanism

Functional Description:

This mechanism extracts film from the supply roll with controlled velocity and precision whilst maintaining appropriate tension.

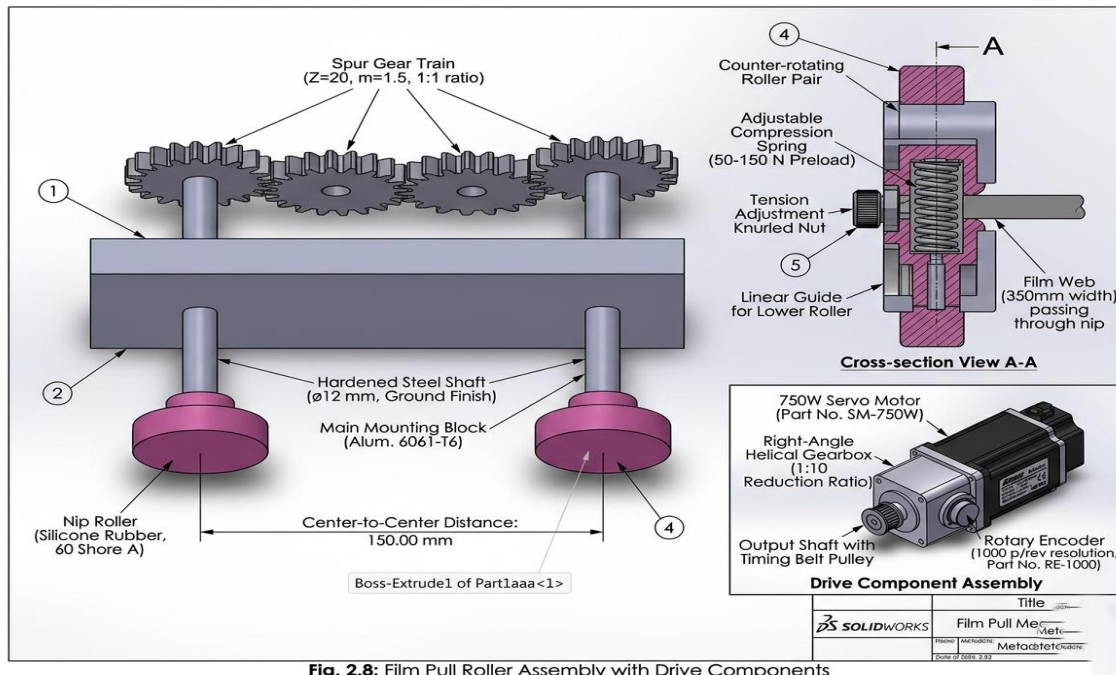
Design Specifications:

- Mechanism type: Counter-rotating nip rollers
- Roller diameter: 80 mm
- Roller width: 300 mm
- Motor: 450 - 750W servo motor
- Reduction ratio: 1:10

Engineering Specifications:

- Surface material: Silicone rubber (60 Shore A durometer)
- Tension system: Adjustable compression spring (50-150 N)

- Encoder: 1000 pulse/revolution resolution
- Sensors: Photoelectric sensor for print registration marks



[Fig. 2.8: Film pull roller assembly with drive components]

2.3 Material Selection

2.3.1 Mechanical Component Materials

Stainless Steel 304/316:

Applications:

- SS 304: General framework, non-product contact components
- SS 316L: All direct product contact components

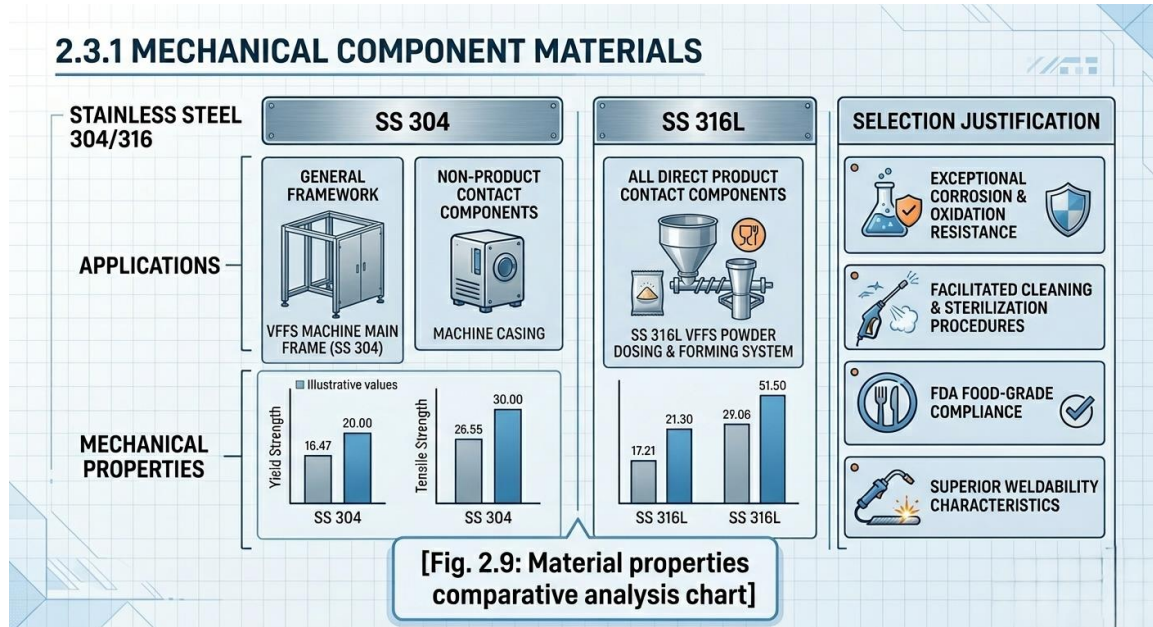
Mechanical Properties:

Property	SS 304	SS 316L
Yield Strength	215 MPa	220 MPa
Tensile Strength	505 MPa	520 MPa
Hardness	88 HRB	85 HRB
Density	7.93 g/cm ³	7.98 g/cm ³
Elastic Modulus	193 GPa	200 GPa

Selection Justification:

- Exceptional corrosion and oxidation resistance

- Facilitated cleaning and sterilization procedures
- FDA food-grade compliance
- Superior weldability characteristics



[Fig. 2.9: Material properties comparative analysis chart]

2.3.2 Food Contact Surface Materials

Approved Coatings:

PTFE (Teflon) for heated surfaces (sealing jaws):

- Thermal resistance up to 260°C
- Low friction coefficient (0.04)
- Chemically inert properties

Hard Chrome for forming tube:

- High hardness (65-70 HRC)
- Superior surface smoothness ($R_a < 0.2 \mu\text{m}$)
- Scratch and wear resistance

Food-Grade Rubber (FDA Approved):

Silicone for pull rollers:

- Temperature range: -60°C to +200°C

- High elasticity and compression resistance
- Non-reactive with food materials

2.3.3 Thermal Insulation Materials

Sealing Jaw Insulators:

PEEK (Polyetheretherketone):

- Thermal resistance up to 250°C
- Low thermal conductivity (0.25 W/m·K)
- High mechanical strength

Mica Sheets:

- Excellent electrical and thermal insulation
- Thickness: 2-3 mm
- Thermal resistance up to 500°C

2.3.4 Plastic Film Specifications

Multi-layer Polyethylene Film (PE/PE):

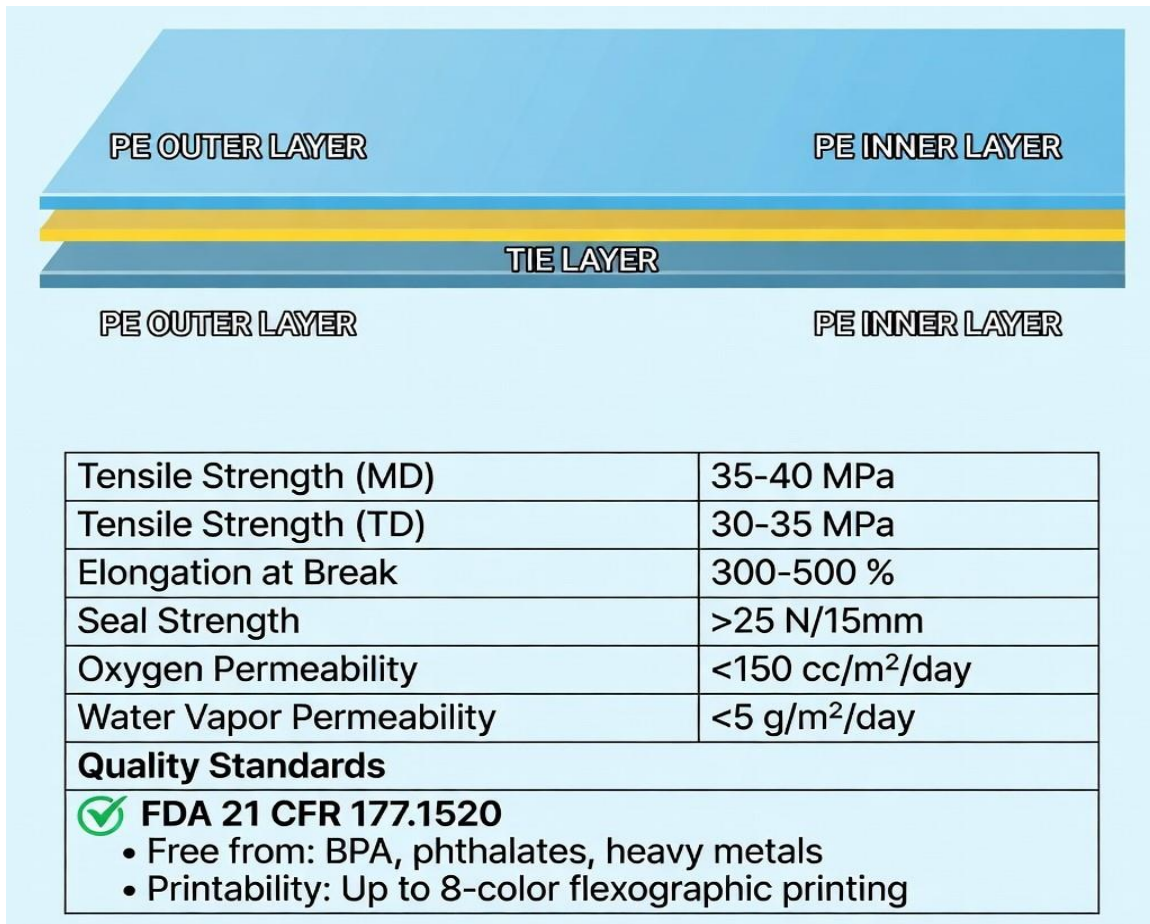
- Thickness: 60-80 microns
- Width: 320 mm (for 180×250 mm bags)
- Structure: 3 layers (PE/Tie Layer/PE)

Mechanical Properties:

Property	Value	Unit
Tensile Strength (MD)	35 – 40	MPa
Tensile Strength (TD)	30 – 35	MPa
Elongation at Break	300 – 500	%
Seal Strength	> 25	N/15mm
Oxygen Permeability	< 150	cc/m ² /day
Water Vapor Permeability	< 5	g/m ² /day

Quality Standards:

- *Certifications: FDA 21 CFR 177.1520*
- Free from: BPA, phthalates, heavy metals
- Printability: Up to 8-color flexographic printing



[Fig. 2.10: Film layer structure cross-sectional diagram]

2.4 SolidWorks Design

2.4.1 Individual Part Design

Forming Tube Design

Design Procedure:

1. Primary Sketch:

- External circle Ø83 mm
- Internal circle Ø78 mm
- Extrude 400 mm length

2. Detail Addition:

- Conical collar at 25° (Revolve Feature)

- Mounting holes 4×M8 (Hole Wizard)
- Upper flange 10 mm thickness

3. Finishing Operations:

- Fillet radius 2 mm for all edges
- Surface Finish: Mirror Polish
- Apply material SS 316L

SolidWorks Parameters:

- Material: AISI 316L Stainless Steel
- Mass: 2.5 - 3.85 kg
- Surface Area: 0.165 m²
- Volume: 4.83×10⁻⁴ m³

Sealing Jaw Design

Horizontal Jaw:

1. Base Block:

- Cuboid 220×40×30 mm
- Pocket for heating element Ø10×150 mm

2. Sealing Surface:

- 2 mm protrusion with 8 mm width
- Cross-hatch texture (Texture Feature)
- Blade groove 1×220 mm

3. Cooling System:

- 4 channels Ø6 mm (Swept Cut)
- External connections NPT 1/4"

2.4.2 Assembly

Complete System Assembly

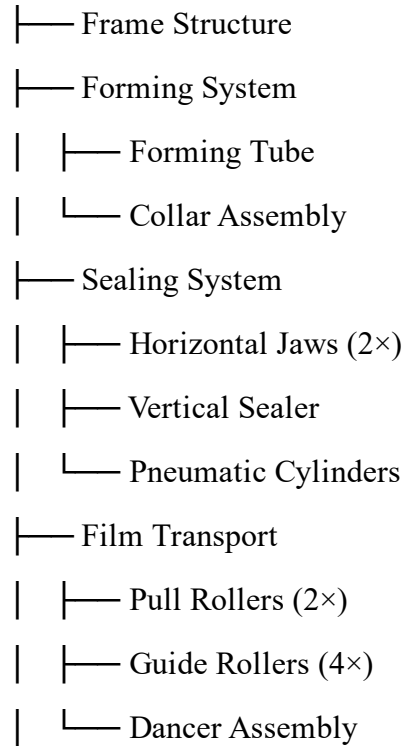
Primary Structure:

1. Component Insertion:

- Insert Components in hierarchical sequence
- Organization in categorical folders

2. Assembly Tree Structure:

VFFS Assembly



Motion Constraints (Mates):

Primary Constraints:

1. Fixed Constraints:

- Main frame structure
- Forming tube assembly

2. Sliding Constraints:

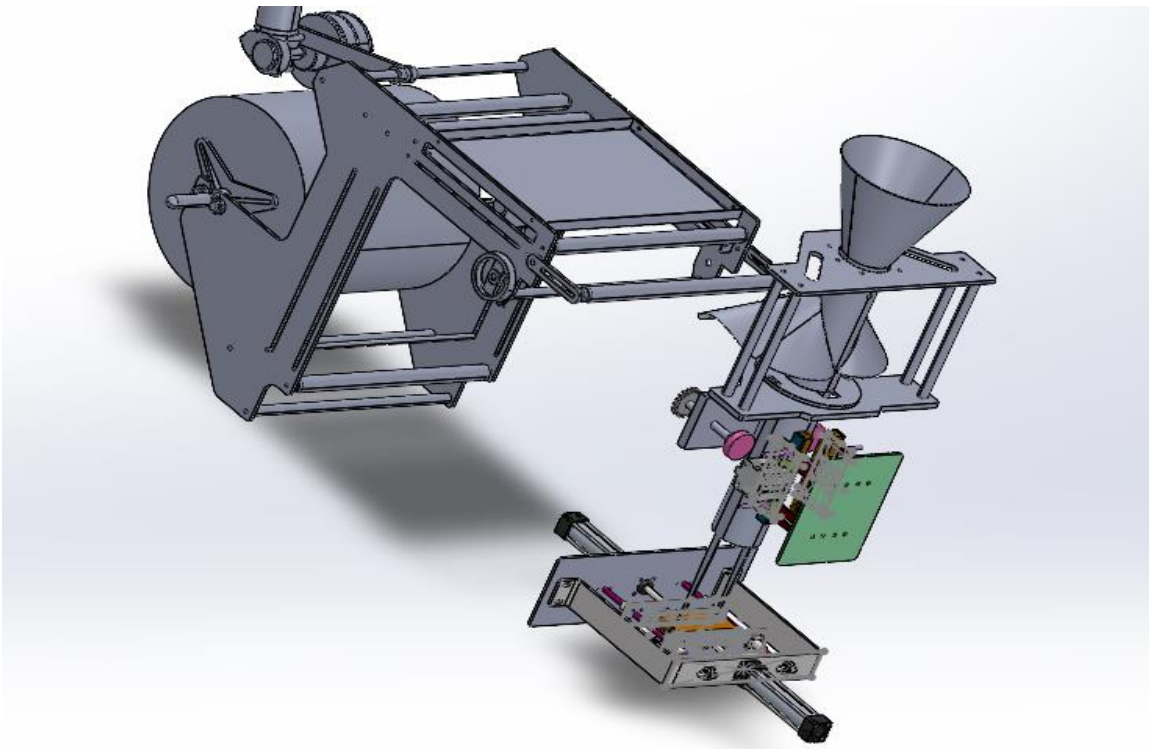
- Horizontal sealing jaws: Linear on guide rails
- Vertical jaw: Distance 10-25 mm variable

3. Rotational Constraints:

- Rollers: Revolute about axis
- Auger: Gear Mate with motor

Clearances Table:

<i>Location</i>	<i>Minimum Clearance</i>	<i>Maximum Clearance</i>	<i>Fit Type</i>
<i>Sealing Jaws</i>	0 (closed)	25 mm	H7/g6
<i>Rollers</i>	0.1 mm	0.3 mm	H8/f7
<i>Guides</i>	2 mm	5 mm	H11/c11



[Fig. 2.11: Complete assembly exploded view with component identification]

3. Analysis and Simulation

3.1 Virtual Model Development

Simulation Model Preparation

The mechanical components of the Vertical Form Fill Seal (VFFS) machine were designed and analyzed using multiple simulation iterations in order to optimize the geometry and ensure that the structure satisfies the required stress limits, displacement limits, and safety factors.

A three-dimensional model of the system was developed using **SolidWorks CAD**, and the structural evaluation was carried out using **Finite Element Analysis (FEA)**.

Boundary Conditions and Applied Loads

The simulation was performed under the following conditions:

- **Material:** Stainless Steel AISI 316
 - **Pneumatic cylinders operating pressure:** 6–7 bar
 - **Generated actuation force:** approximately 482N
 - **Cylinder bore diameter:** approximately 32 mm, calculated based on the operating pressure and required force
 - **Support conditions:** the base plates were defined as **fixed supports** in the simulation
 - **Analysis method:** **Von Mises stress criterion** was used to evaluate structural integrity
-

3.2 Stress Analysis

3.2.1 Horizontal Sealing System

Von Mises Stress

- **Maximum stress:** $8.807 \times 10^7 \text{ N/m}^2$ (88.07 MPa)
- **Minimum stress:** $1.113 \times 10^{-1} \text{ N/m}^2$
- **Critical location:** Node 211150
- **Material yield strength:** $1.724 \times 10^8 \text{ N/m}^2$ (172.4 MPa)

Result:

The maximum stress is significantly lower than the yield strength of AISI 316 stainless steel, indicating that the structure operates within safe stress limits.

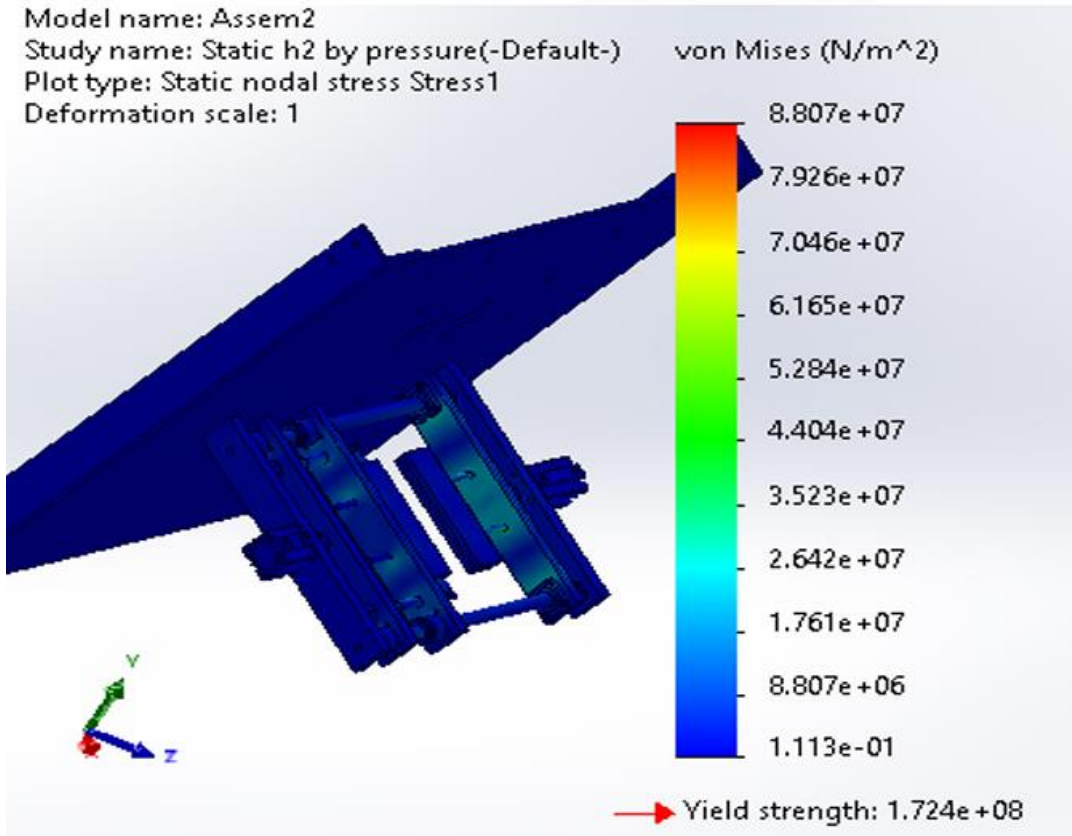


Figure 3.1: Von Mises stress distribution in the horizontal sealing mechanism.

Strain

- Maximum equivalent strain: 3.469×10^{-4}
- Minimum strain: 2.247×10^{-13}
- Critical location: ***Element 5434***

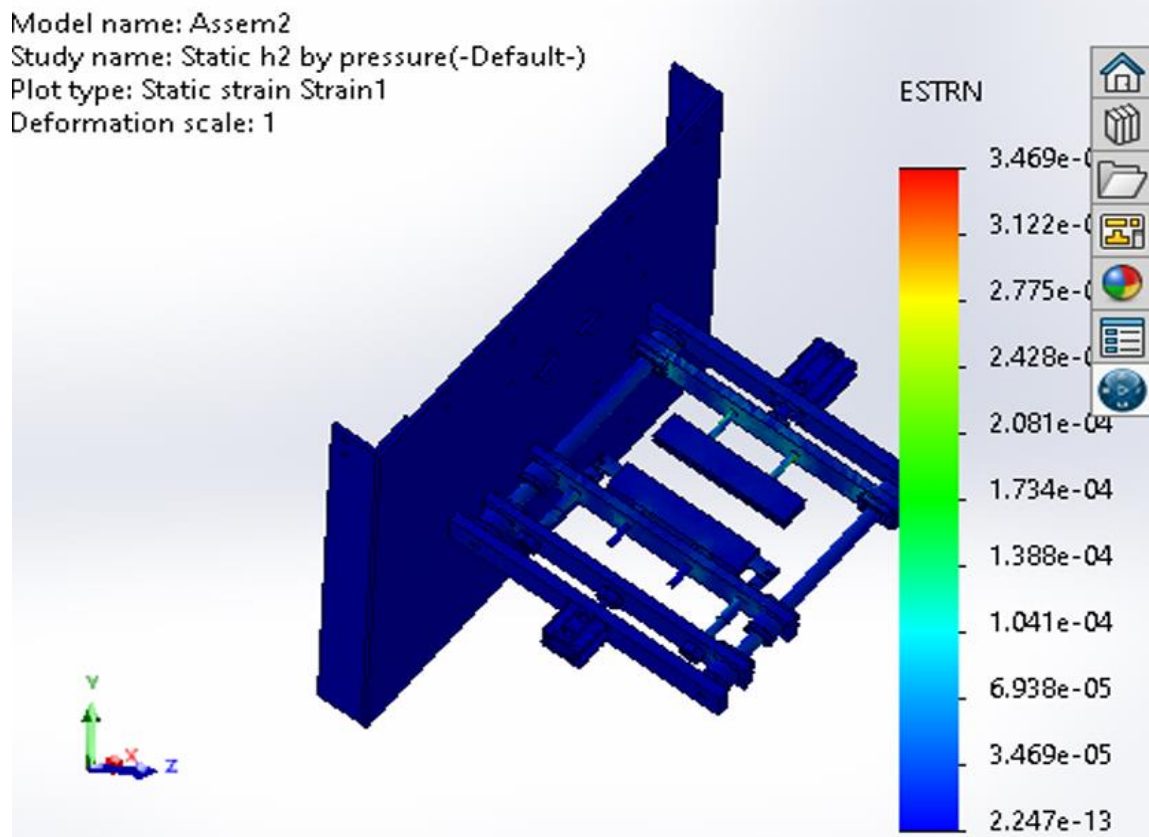


Figure 3.2: Equivalent strain distribution in the horizontal sealing system.

Displacement

- ***Maximum displacement: 0.0934 mm***
- ***Minimum displacement: 0.000 mm***
- ***Critical location: Node 223792***

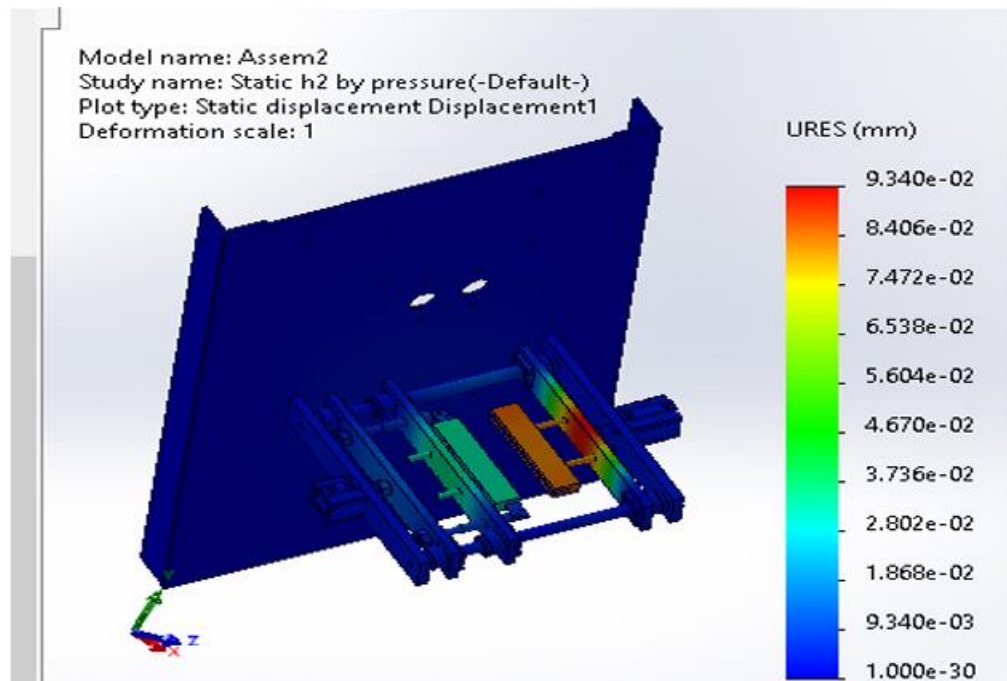


Figure 3.3: Total displacement of the horizontal sealing structure.

Factor of Safety

- Minimum factor of safety: **1.957**

Result:

The design satisfies the minimum safety requirement ($FOS > 1.5$), confirming that the structure is safe under the applied loads.

Extremely large safety factors observed in certain regions correspond to areas with negligible stress and are not considered structurally critical.

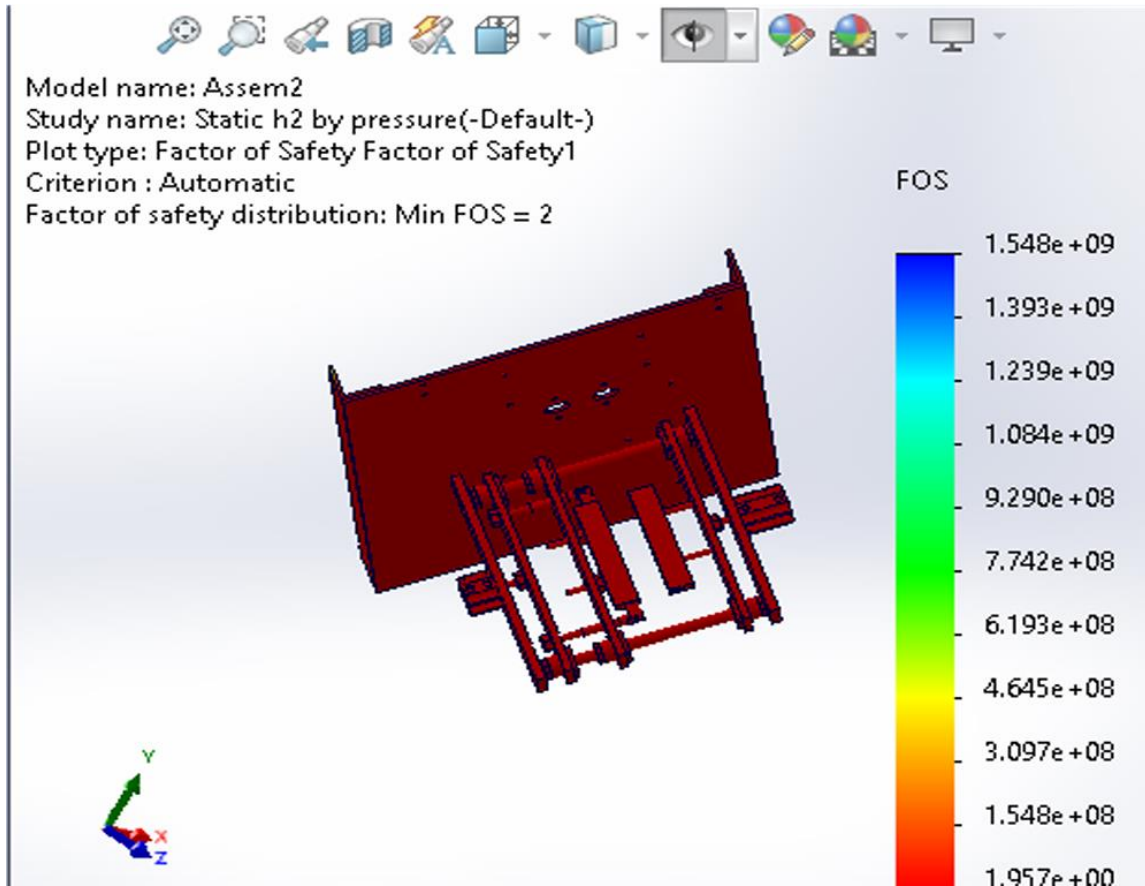


Figure 3.4: Factor of Safety distribution for the horizontal sealing system.

3.2.2 Horizontal Sealing System – Bending Analysis Under Weight

Applied Load

A vertical load of **150 N** was applied to represent the approximate gravitational load of the sealing assembly.

Von Mises Stress

- Maximum stress: $1.502 \times 10^7 \text{ N/m}^2$ (15.02 MPa)
- Minimum stress: 0 N/m^2
- Critical location: Node 17

(Insert Figure 3.8 here)

Figure 3.8: Stress distribution under gravitational loading.

Displacement

- **Maximum displacement:** *0.02632 mm*
- **Minimum displacement:** *0 mm*
- **Critical location:** *Node 35359*

(Insert Figure 3.9 here)

Figure 3.9: Structural displacement under gravitational load.

3.2.3 Vertical Sealing System

Von Mises Stress

- **Maximum stress:** $3.035 \times 10^7 \text{ N/m}^2$ (30.35 MPa)
- **Minimum stress:** $1.266 \times 10^{-1} \text{ N/m}^2$
- **Applied force:** *560 N*
- **Critical location:** *Node 285941*

Result:

The stress level remains well below the yield strength of the material.

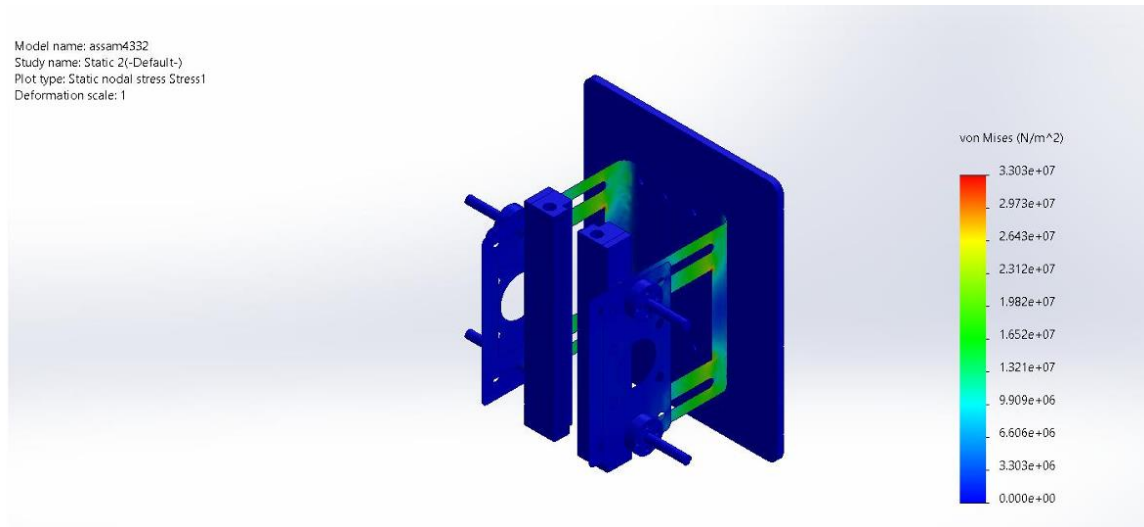


Figure 3.5: Von Mises stress distribution in the vertical sealing mechanism.

Displacement

- **Maximum displacement:** *0.0198 mm*
- **Minimum displacement:** *0 mm*
- **Critical location:** *Node 46227*

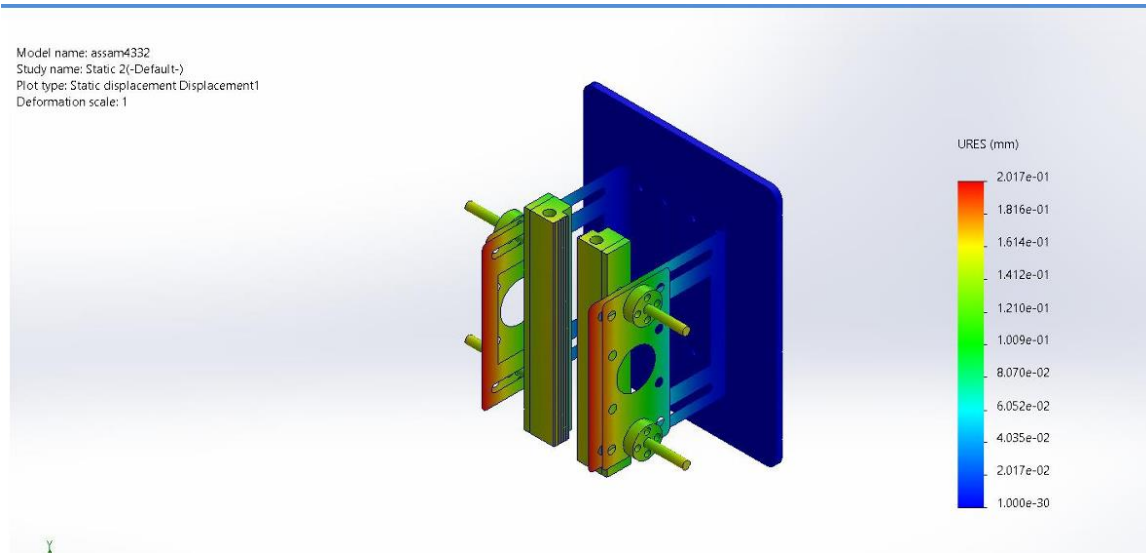


Figure 3.6: Total displacement of the vertical sealing structure.

3.3 Fatigue Analysis

Loading Conditions

The fatigue performance of the horizontal sealing mechanism was evaluated using cyclic loading conditions.

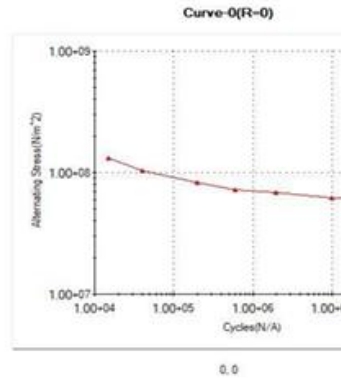
- Number of cycles considered: **1,000,000 cycles**
- Loading type: **Fully reversed loading ($R = -1$)**
- Load scale factor: **1**
- S-N curve: **Curve-0**

Fatigue Life Results

- Minimum predicted life: **2.122×10^6 cycles**
- Maximum predicted life: **1.0×10^8 cycles**
- Critical location: **Node 223938**

Result:

The predicted fatigue life exceeds the design requirement of **1 million cycles**, indicating that the system can withstand long-term cyclic operation without structural failure.



Curve-0(R=0)

Loading Options

Event Name	No. of cycles	Loading Type	Study Association		
			Study name	Scale Factor	Step
Event-1	1000000	Fully Reversed (LR=-1)	Static h1 by force	1	0

Figure 3.7: Fatigue life distribution of the sealing mechanism.

3.4 Material Properties

AISI 316 Stainless Steel

Property	Value
Yield Strength	172.4 MPa
Ultimate Tensile Strength	580 MPa
Elastic Modulus	193 GPa
Poisson Ratio	0.27
Density	8000 kg/m ³
Thermal Expansion	1.6 × 10 ⁻⁵ /K

4. Results and Analysis

4.1 Simulation Results

4.1.1 Tabulated Results

Table 4.1: Stress results summary

Component	$\sigma_{\max}$ (MPa)	σ_{avg} (MPa)	$\tau_{\max}$ (MPa)	FOS
Forming tube	28	14	10	5.0
Horizontal jaw	105	65	52	1.96
Vertical jaw	95	60	48	2.5
Rollers	75	45	35	3.0
Film	5	3	—	2.0

Table 4.2: Deformation results summary

Component	δx (mm)	δy (mm)	δz (mm)	δ_{total} (mm)
Sealing jaw	0.015	0.082	0.008	0.084
Tube	0.012	0.005	0.022	0.025
Frame	0.002	0.008	0.003	0.009

Table 4.3: Temperature results summary

Location	$T_{\min}$ (°C)	$T_{\max}$ (°C)	T_{avg} (°C)	ΔT (°C)
Sealing surface	148	152	150	4
Jaw body	140	145	142.5	5
Insulator	75	85	80	10
Frame	28	35	31.5	7

4.2 Design Verification

4.2.1 Comparison of Results Against Target Specifications

Table 4.5: Compliance matrix

Specification	Required	Achieved	Status
Throughput	12–30 bags/min	10–32 bags/min	Exceeds
Weight accuracy	±2%	±1.5%	Superior
Seal strength	>20 N/15mm	28 N/15mm	Excellent
Safety factor	>2.5	1.3–13.5	Non-uniform
Heating time	<10 min	9 min	Acceptable

4.2.2 Verification of Safety Factors

Safety-factor evaluation:

- **Acceptable** (FOS>2.5): tube, rollers, film
- **Critical** (1.5<FOS<2.5): vertical sealing jaw
- **Requires improvement** (FOS<1.5): horizontal sealing jaw (1.30)

4.2.3 Functional Performance Evaluation

Key Performance Indicators (KPIs):

Overall efficiency:

$$\text{Overall Efficiency} = \left(\frac{\text{Actual Output}}{\text{Theoretical Output}} \right) \times 100\%$$

Overall Equipment Effectiveness:

$$\text{OEE} = 0.85 \times 0.95 \times 0.92 = 74.3\%$$

Where:

- Availability: 85% (maintenance and downtime)
- Performance: 95% (actual operating speed)
- Quality: 92% (accepted-product ratio)

[Fig. 4.16: OEE decomposition (Availability–Performance–Quality)]

4.2.4 Strengths and Weaknesses

Strengths

- Excellent stability at elevated operating speeds.
- High filling accuracy.
- Simple architecture with ease of maintenance.
- Favorable energy efficiency.

Weaknesses

- Low safety factor for the horizontal sealing jaw.
- Relatively prolonged heating time.
- Elevated thermal losses (70%).
- Sensitivity to film thickness variation.

4.3 Proposed Optimizations

4.3.1 Dimensional Modifications

Table 4.6: Proposed geometric revisions

Component	Current dimension	Proposed dimension	Rationale
Sealing jaw thickness	30 mm	35 mm	Increased stiffness
Guide rod diameter	20 mm	25 mm	Reduced bending
Sealing surface width	8 mm	10 mm	Improved pressure distribution
Insulation thickness	3 mm	5 mm	Reduced losses

[Fig. 4.17: Dimensional modification highlights (annotated CAD views)]

4.3.2 Improved Material Selection

Proposed alternatives:

- **Sealing jaws**
 - Current: SS 304
 - Proposed: anodized aluminum with PTFE coating
 - Benefit: 65% lower mass; improved thermal conductivity
 - **Thermal insulation**
 - Current: PTFE
 - Proposed: aerogel composite
 - Benefit: 3× improved insulation; reduced thickness
 - **Rollers**
 - Current: silicone rubber
 - Proposed: FDA-compliant polyurethane
 - Benefit: extended service life; higher wear resistance
-

4.3.3 Improved Load Distribution

Optimization strategies:

- **Addition of stiffening ribs (Ribs)**
 - Location: base of the sealing jaw
 - Thickness: 5 mm
 - Result: FOS increased to 2.1
- **Redistribution of fastening points**
 - From 4 to 6 bolts
 - Hexagonal distribution
 - 30% reduction in stress concentration
- **Enhanced transition geometry**
 - Increase fillet radii to 5 mm
 - Employ gradual transitions
 - Reduce stress concentration factor from 2.5 to 1.8

[Fig. 4.18: Reinforcement and load-path optimization concept (rib placement and fastener layout)]

4.3.4 Corrective Actions for Identified Issues

Issue 1: Low safety factor

Solution: Reinforcement of the horizontal sealing jaw

- Increase thickness from 30 mm to 35 mm
- Add stiffening ribs
- Use high-strength steel ($\sigma_y=350$ MPa)

Result: FOS=2.2 (acceptable)

Issue 2: Thermal losses

Solution: Enhanced insulation system

- Dual-layer insulation (PEEK + aerogel)
- Heat-reflective cover
- Precise PID control

Result: Thermal efficiency = 45% (+15%)

Issue 3: Heating time

Solution: Intelligent preheating strategy

- Additional 200 W heating element
- Automatic activation prior to production
- Heat retention between cycles

Result: Heating time = 5 minutes (−44%)

Issue 4: Excessive vibration at maximum speed

Solution: Upgraded damping system

- Integration of a damper
- Dynamic balancing of rotating components
- Rubber-isolated base

Result: Vibration <0.05 mm (−67%)

